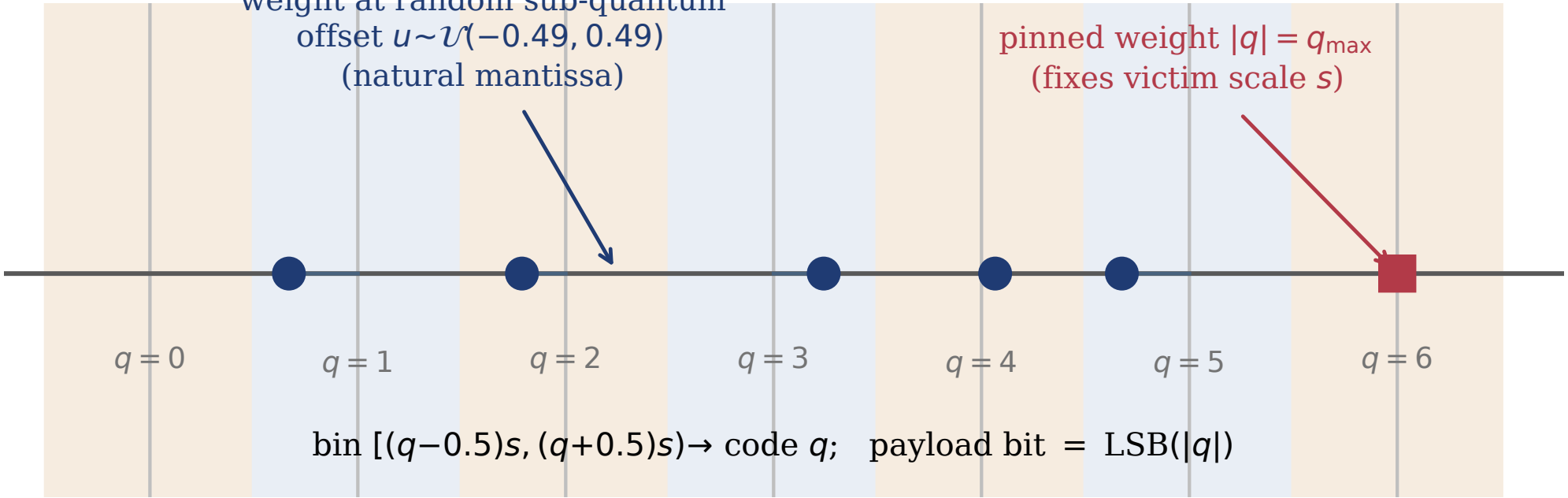

weight at random sub-quantum
offset $u \sim \mathcal{U}(-0.49, 0.49)$
(natural mantissa)
pinned weight $|q| = q_{\max}$
(fixes victim scale s)
$q = 0$
$q = 1$
$q = 2$
$q = 3$
$q = 4$
$q = 5$
$q = 6$
bin $[(q-0.5)s, (q+0.5)s) \rightarrow$ code q; payload bit = LSB($|q|$)